Rylan Polster

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EDUCATION

Yale University — 2021–Present

B.S. Electrical Engineering and Computer Science, Expected 2025

- Cumulative GPA: 3.9
- Relevant coursework: computer architecure, VLSI design, compilers, systems programming, electronic circuits, computer graphics/vision, data structures, algorithms, and physics
- Capstone in progress: Program a compiler to generate CMOS circuits that implement asynchronous algorithms
- Extracurricular groups: Yale Dramatic Association, Yale Computer Society, Yale IEEE Chapter
- Selected student employment: Event Technician, Undergraduate Production Peer Mentor*

**More information available upon request*

EXPERIENCE

Homebrew — Package Manager for macOS and Linux — 2020–Present

Technical Steering Committee Member (Appointed) and Maintainer — Remote

- Top 10 committer with over 2,000 commits related to user experience, efficiency, and bug fixes
- Programmed the largest internal functionality update since 2016, resulting in drastically faster operation
- Strategize technical goals for the organization and rule on technical disputes between maintainers
- Triage, review, and merge over 2,000 pull requests and issues submitted by Homebrew's millions of users

Asynchronous VLSI Chip Tapeout — Ongoing Project — 2024–Present

Designer — New Haven, CT

- Design, layout, and simulate transistor networks for an asynchronous ASIC to be manufactured by TSMC
- Work with the SkyWater 130nm Production PDK (for testing) and TSMC's 180nm proprietary design rules

Yale Dramatic Association — 501(c)(3) Production Company — 2023–2024

Production Manager and Executive Board Member — New Haven, CT

- Oversaw all use of association wood shop, and managed equipment training and maintenance
- Organized and led all construction/installation of scenery, lights, and sound equipment for 6 show season
- Approved professional & student designs across 6 departments to fit time, labor, budget, and safety constraints
- Authored technical planning documents and schematics using Vectorworks and other CAD software

Smartphone Biopsy Needle Guidance Device Project — 2021–2022

Software Developer — Cleveland, OH

- Developed a smartphone application to improve accuracy and speed in CT-guided needle interventions
- Co-authored a study published in Skeletal Radiology demonstrating the benefit of using the application which was presented at the Society of Skeletal Radiology's 2023 annual meeting

CWRU Institute for Smart, Secure, and Connected Systems — 2019–2020

Software Engineering Intern — Cleveland, OH

- Developed an IoT device to monitor and report wheel wear on local transit cars
- Prototyped a LoRaWAN system for monitoring industrial power consumption

PUBLICATIONS

Lui, C., **Polster, R.**, Bullen, J. *et al.* Smartphone application with 3D-printed needle guide for faster and more accurate CT-guided interventions in a phantom. *Skeletal Radiology* (2023). 10.1007/s00256-023-04453-x

SKILLS AND TECHNOLOGIES

- VLSI: Magic VLSI, IRSim, Xyce, ACT Asynchronous VLSI Flow
- Software Engineering: C, C++, Ruby, Python, Bash, Git, GitHub Actions, AWS, MQTT, LoRaWAN, Arduino